Gateway API Inference Extension

Generative AI Inference on Kubernetes

🏷️ Kubernetes 🏷️ Networking 🏷️ Machine Learning

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Topics

- **Gateway API**
- **Ingress**
- **Gateway API Inference Extension**
- **마치며**

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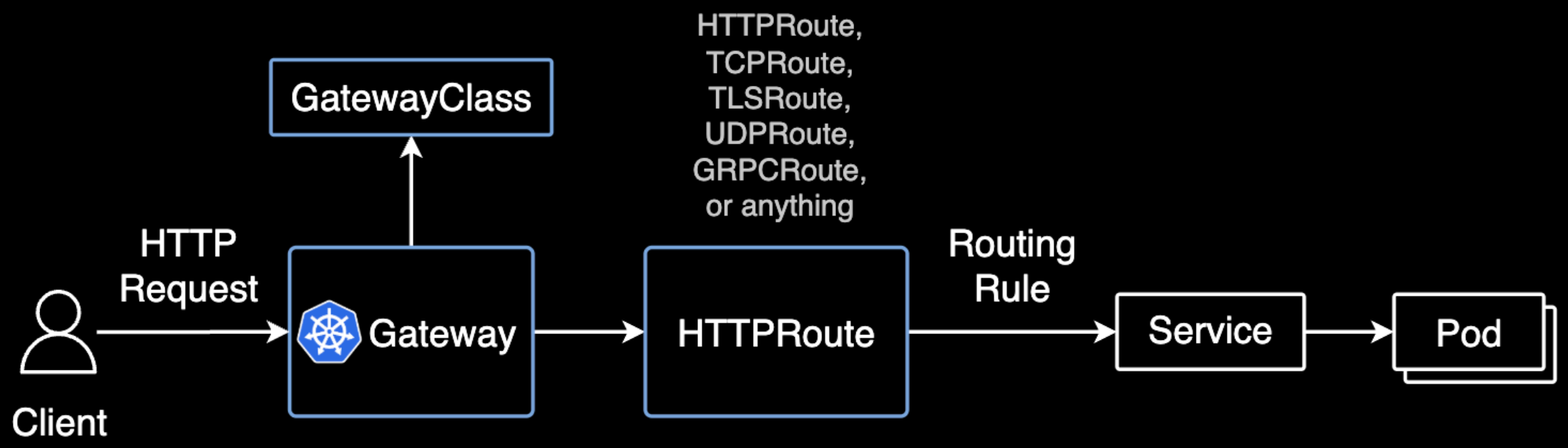
Gateway API



Gateway API is an official Kubernetes project focused on L4 and L7 routing in Kubernetes

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Ingress

쿠버네티스 공식문서에 의하면 Ingress는 더 이상 업데이트되지 않고 Gateway API에 새로운 기능이 추가되고 있습니다.

> Ingress is frozen. New features are being added to the Gateway API.

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Gateway API Inference Extension

gateway api
inference extension

Gateway API의 확장 기능인 Inference Extension이 공식적으로 출시되었음

News: Google, Bytedance, and Red Hat make Kubernetes generative AI inference aware

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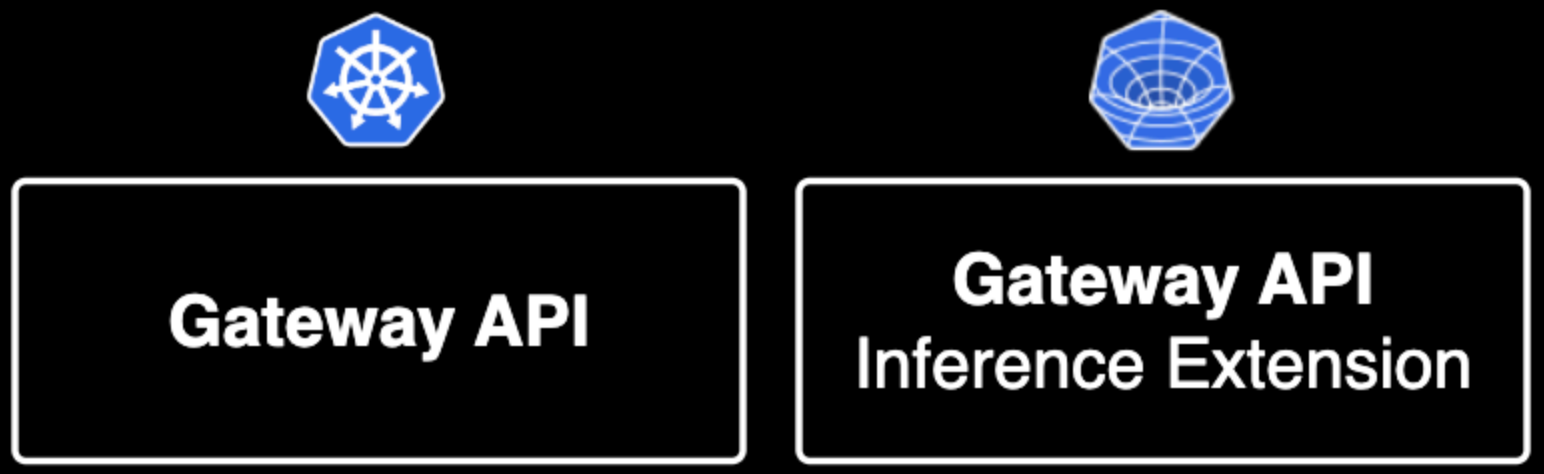
Gateway API vs Inference Extension



Inference Extension은 기존 Gateway API의 기능을 확장해서 추론 모델의 라우팅을 지원함.

자세한 핵심 기능들은 Key Features에서 확인할 수 있음.

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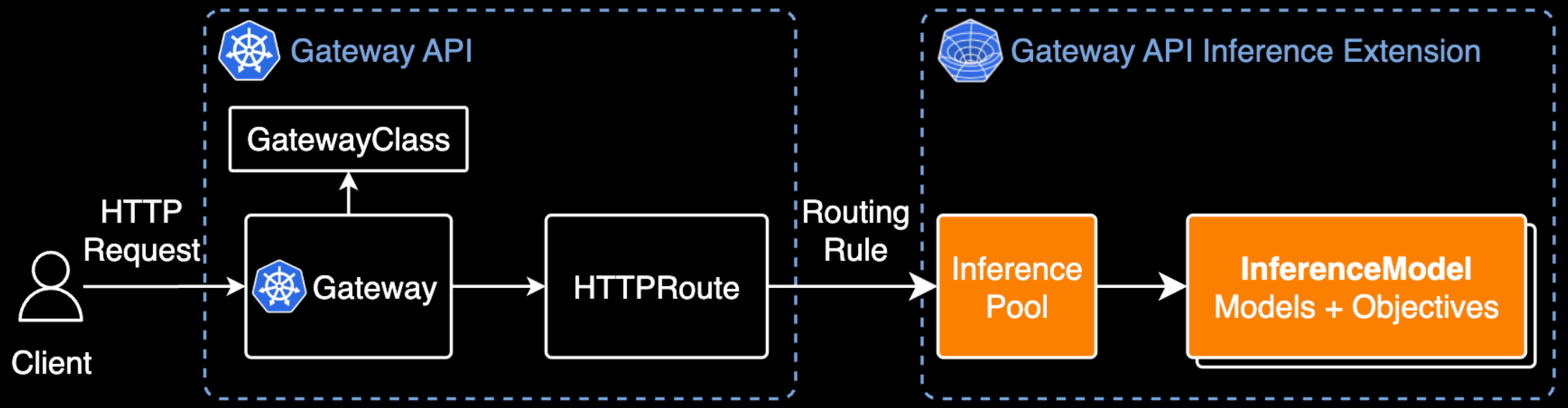
Areas of Interest



an inference gateway - supporting inference platform teams self-hosting large language models on Kubernetes.

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마치며

Networking API의 현상황

- 쿠버네티스 로드맵에 따라 여러 인그레스 컨트롤러와 서비스메시 도구들이 자신들의 커스텀 리소스 대신 Gateway API를 표준으로 사용하는 움직임을 보임. 대표적으로 Istio도 Gateway API를 적극 채택함.
- Ingress 리소스는 여전히 사용되지만 더 이상 업데이트되지 않는(frozen) 상태이고 Gateway API에만 새로운 기능이 추가되고 있다는 점을 명심해야 함.

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References

- Gateway API
- Gateway API FAQ
- Gateway API Inference Extension
- News: Google, Bytedance, and Red Hat make Kubernetes generative AI inference aware
- Istio Gateway API
- Ingress + API Gateway = Kubernetes Gateway API

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